

Foundations for programming languages 1 --- Supplement

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References

Programming Languages Concepts & Constructs
2nd edition, Ravi Sethi, Addison-Wesley, 1996.

Concepts in Programming Languages,
John C. Mitchell, Cambridge University Press, 2001.

Supplement 1

The definition of syntax of the Little Quilt language

- allows e_1 and e_2 to have different height in $\text{sew}(e_1, e_2)$.
- allows the use of undefined variables.
- allows the mismatch between the number of actual arguments and formal parameters in function applications.

This kind of things are not checked in syntax analysis. Instead, they are checked in some phase after the parsing or in runtime.

Supplement 2

The built-in function of the language Little Quilt is *prefix*, while some other constructs may be *infix*, as is often the case with arithmetic operators like + in 1+2.

Grammar classification by Noam Chomsky

Type 0: grammars with no restriction

These generate any language that can be recognized by a Turing machine.

Type 1: context sensitive grammars

Type 2: context free grammars

Type 3: regular grammars